Swathi Shree Narashiman

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EDUCATION

- **Stanford University, California, USA** : *Sep '26 – Jun '28*
  Masters in Electrical Engineering | AI focus
  Current coursework: *Decision Making Under Uncertainty; Efficient Generative Models; Statistical Inference*
- **Indian Institute of Technology Madras** : *Nov '22 – May '26*
  B.Tech Electrical Engineering (Hons.) & Minor in Computing **CGPA:** 9.64/10

WORK EXPERIENCE

- **Fujitsu Research** *Jun '26 – Aug '26*
  Research Intern, AI Lab (LLM Agents & Agentic AI) *Bengaluru, IN*
  - Developed a **workflow-centric procedural memory** framework that abstracts trajectories into reusable workflows.
  - Reduced memory footprint by **3.5**× via online workflow construction & retrieval with competitive performance.
  - Work accepted to the 2nd Workshop for Research on Agent Language Models (**REALM**) at **EMNLP 2026**.

RESEARCH EXPERIENCE

- **Summer@EPFL Research Internships** *May '24 – Jul '25*
  Mentors: Prof. Nicholas Flammarion (TML Lab) & Prof. Haitham Al Hassanieh (SENS Lab) *Lausanne, CH*
  - Selected twice for the competitive Summer@EPFL research program among the **top 1.6% globally**.
  - Analyzed **Induction heads** in transformers, demonstrating N-gram interpolation; Work published at **ICML 2026**.
  - Developed Point cloud based methods for **3D radar shape reconstruction**, published as a **BMVC 2025** paper.
- **Bachelor Thesis** *Jan '26 – May '26*
  Mentor: Prof. Andrew Thangaraj, IIT Madras *Chennai, India*
  - Proposed embedding-based **Semantic Smoothing** for Distribution Estimation with Minimax-optimal guarantees.
  - Improved perplexity by 28% over Add-Constant and 20% over Kneser-Ney; Work accepted at the **ITW 2026**.
- **Theoretical Analysis of Large Language Models** *Jan '25 – May'26*
  Mentor: Prof. Nambi Seshadri, UC San Diego *Remote*
  - Characterized a Log-linear law in the ordered logit spectrum of Autoregressive LMs, observing consistent scaling across architectures and model sizes. Work accepted in the **PriGM** Workshop at **Neurips 2026**.
- **Music Interpretability via Activation Steering** *Apr '24 – May '25*
  Club: AI Club, Center for Innovation, IIT Madras *Chennai, India*
  - Developed **activation steering** for controllable genre, instrument & pitch transfer in MusicGen; published on arXiv.

PUBLICATIONS

1. **S. S. Narashiman**, A. Ramana V, N. Seshadri, *The Geometry of Language Model Logits: Generic Structure and Learned Departures*, PriGM Workshop @ NeurIPS 2026.
2. **S. S. Narashiman**, I. Jindal, M. Chandran, *From Trajectories to Workflows: Online Procedural Memory Abstraction for Language Agents*, REALM Workshop @ EMNLP 2026.
3. F. D'Angelo, O. K. Yuksel, **S. S. Narashiman**, et al., *Induction Heads Interpolate N-Grams*, ICML 2026.
4. H. Balasundaram, **S. S. Narashiman**, P. Mathur, et al., *Semantic Smoothing for Language Models via Distribution Estimation and Embeddings*, IEEE ITW 2026.
5. S. Hussein, J. Guan, **S. S. Narashiman**, et al., *3D Shape Reconstruction from Autonomous Driving Radars*, BMVC 2025.

RELEVANT UNDERGRADUATE COURSEWORK

- **Machine Learning & AI :** Advanced Topics in AI; Stochastic Modeling & Queues; Natural Language Processing
- **Algorithms & Computing :** Randomized Algorithms; Quantum Algorithms; Computer Organization; Cryptography
- **Mathematics & Theoretical Foundations :** Linear Algebra; Convex Optimization; Incentive Centered Design (Game Theory); Probability; Control Systems; Information Theory; Multivariable Calculus; Series & Matrices

SKILLS

Python, C/C++, PyTorch, JAX, CUDA, Hugging Face Transformers, vLLM, LangChain, Docker, Git, RunAI